

TCS NQT PYQ QUESTIONS

Company: TCS NQT

Round: Online

Year: 2025

Difficulty: Medium-Hard

Topic: Speed, Time and Distance (Trains)

Problem Description:

Two trains, 150 m and 120 m long, are running on parallel tracks in opposite directions at speeds of 54 km/h and 36 km/h respectively.

Find the time taken by the two trains to completely cross each other.

Options:

- A) 9 seconds
- B) 10 seconds
- C) 10.8 seconds
- D) 12 seconds

Correct Answer:

C) 10.8 seconds

Approach to Solve This Problem:

STEP 1:

Since the trains move in opposite directions, their relative speed is the sum of their individual speeds: $54 + 36 = 90$ km/h.

Convert to m/s: $90 \times (5/18) = 25$ m/s.

STEP 2:

To completely cross each other, the trains together must cover a distance equal to the sum of their lengths: $150 + 120 = 270$ m.

STEP 3:

Time taken = Distance / Relative Speed = $270 / 25 = 10.8$ seconds.

CORRECT ANS = 10.8 seconds